FSOT Mathematical Key — One Page for Scientists

Fluid Spacetime Omni-Theory · pin **D1D38A** · zero free parameters · github.com/dappalumbo91/FSOT-2.1-Lean · full key: docs/FSOT_MATH_KEY.md

1. Unified principle

One seed-derived scalar engine evaluated at preregistered dimensional interfaces (D_{eff}). Seeds π, e, ϕ, γ, Catalan G only — **no free fit parameters**. Domains are folds of the same engine, not separate fitted models. When residuals fail: fix D_{eff} / observer / C_{factor} / Poof path — do not invent a new coefficient.

Scalar: $S = K\cdot(T_1+T_2+T_3)$ · T_1 observer-modulated base (consciousness factor C_{factor} when observed) · T_2 linear · T_3 valve–acoustic–phase (Poof, Suction, Chaos, bleed).

Prediction law (every domain)

```
computed = measured × (1 + |S(domain)| × factor)
```

Implementation: vendor/fsot_compute.py · scripts/fsot_api_predict_lib.py (domain_scalar, fsot_scaled, PROPERTY_ROUTING).

2. Recipe — use the math in any domain

Step	Action
1	Name measured m with public/lab provenance
2	Pick domain / D_{eff} (micro 5–9 · meso 10–15 · geo 16–19 · astro 20–25)
3	S = domain_scalar(name) — full stack at that interface
4	computed, err% = fsot_scaled(m, name) or make_fsot_record(...)
5	Green if domain median residual ≤ **0.5%** (aspiration ≤ 0.05%)
6	Optional formal lock: export gate → Lean / Coq / Isabelle / SMT / TLA+

3. Snapshot (this repository edition)

Quantity	Value	Quantity	Value
Atlas domains	403	Green benchmarks	432/432
Empirical records (atlas)	61,335 (envelope)	Catalog obligations	2025
MPCORB objects	1,554,101	MPCORB pooled residual	0.023%

4. Verification stack (layers A / B / C)

A Engine math: Lean 4 master · Coq · Isabelle · F* · Rust replay — seeds, raw_S, exported bounds. **B Empirical:** domain residual ledger ≤ 0.5% green. **C Streams/catalogs:** live APIs, MPCORB Kepler integrity, MAST, etc. **Bulk:** SMT Z3/CVC5 residual conjunction. **Flow:** TLA+ domain-routing (no gate skips). Multi-prover locks *exported gate literals*; it does not re-derive raw telescope pixels in type theory.

5. Reproduce

git clone https://github.com/dappalumbo91/FSOT-2.1-Lean.git && python scripts/run_publication_verification_bundle.py
MPCORB: python scripts/ingest_mpcorb_catalog.py && python scripts/build_mpcorb_fsot_benchmark.py
Margin: python scripts/audit_all_benchmark_margins.py → data/benchmark_margin_audit.json

One seed set · one scalar S · one prediction law · many D_{eff} interfaces. That is the mathematical key for every FSOT domain.